Computer Networks - *Xarxes de Computadors*

Outline

- Course Syllabus
- Unit 1: Introduction
- **Unit 2. IP Networks**
- Unit 3. TCP
- Unit 4. LANs
- Unit 5. Network applications

Unit 2: IP Networks

Outline

- **IP layer service**
- IP addresses
- Subnetting
- Routing tables
- ARP protocol
- IP header
- ICMP protocol
- DHCP protocol
- Routing algorithms
- NAT
- Security in IP

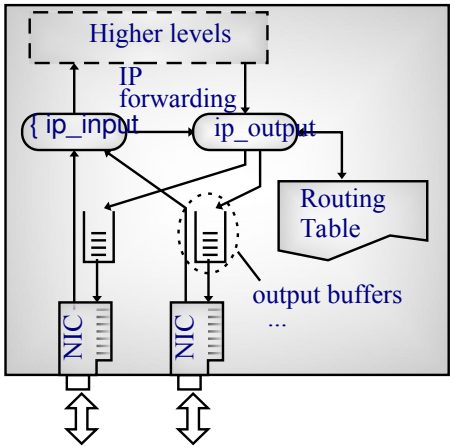
Unit 2: IP Networks

IP Layer Service

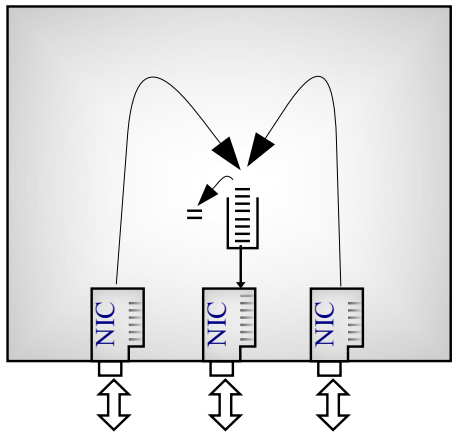
- Internet Protocol (IP) goal is routing datagrams.
- IP main design goal was interconnecting hosts attached to LANs/WANs networks of different technologies.
- IP characteristics are:
  - Connectionless
  - Stateless
  - Best effort



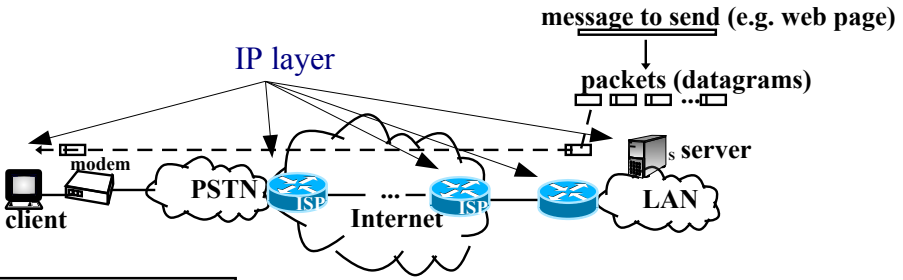
Commercial routers
(edge routers)



Basic router architecture



Looses may occur due to buffer overflow



Unit 2: IP Networks

High Performance Routers

(core routers)



Source: Juniper

"There is a major upgrade going on in service providers upgrading their core networks," Chris Komatas, director of service provider marketing at Juniper, said.

"The next-generation core network is all about having the agility to support any service. T1600 is delivering No. 1 in scale, No. 1 in service control and No. 1 in efficiency. All the metrics that are important for a service provider."

The keys to the performance throughput on the Juniper T1600 are the 100Gbps-capable slots that can support all the major connectivity options that carriers may have. Among them is support for OC-768 (40 Gbps), OC-192 (10Gbps) and 10GbE (10 Gigabit Ethernet).

Juniper (www.juniper.net)

Figure 1. Cisco XR 12000 and 12000 series routing portfolio



Table 1. Product Specifications

Product Specification	Cisco XR 12000 and 12000 Series 16-Slot Chassis	Cisco XR 12000 and 12000 Series 10-Slot Chassis	Cisco XR 12000 and 12000 Series 6-Slot Chassis	Cisco XR 12000 and 12000 Series 4-Slot Chassis
Slot capacity	16 slots	10 slots	6 slots	4 slots
Aggregate switching capacity	Cisco 12016: 80 Gbps Cisco 12416: 320 Gbps Cisco 12816: 1280 Gbps	Cisco 12010: 50 Gbps Cisco 12410: 200 Gbps Cisco12810: 800 Gbps	Cisco 12006: 30 Gbps Cisco 12406: 120 Gbps	Cisco 12404: 80 Gbps
Full-duplex throughput per slot	Cisco 12016: 2.5 Gbps/slot Cisco 12416: 10 Gbps/slot Cisco 12816: 40 Gbps/slot	Cisco 12010: 2.5 Gbps/slot Cisco 12410: 10 Gbps/slot Cisco 12810: 40 Gbps/slot	Cisco 12006: 2.5 Gbps/slot Cisco 12406: 10 Gbps/slot	Cisco 12404: 10 Gbps/slot

cisco (www.cisco.com)

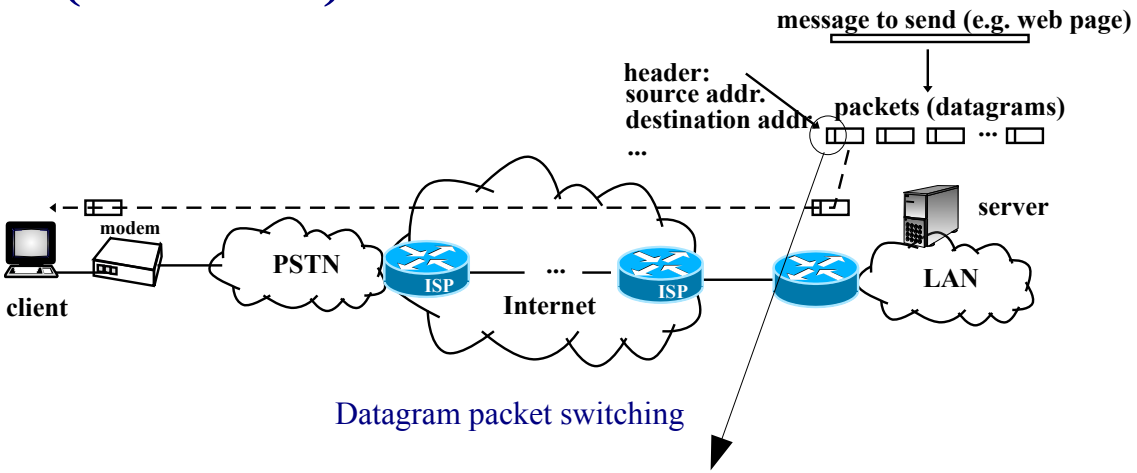
Unit 2: IP Networks

Outline

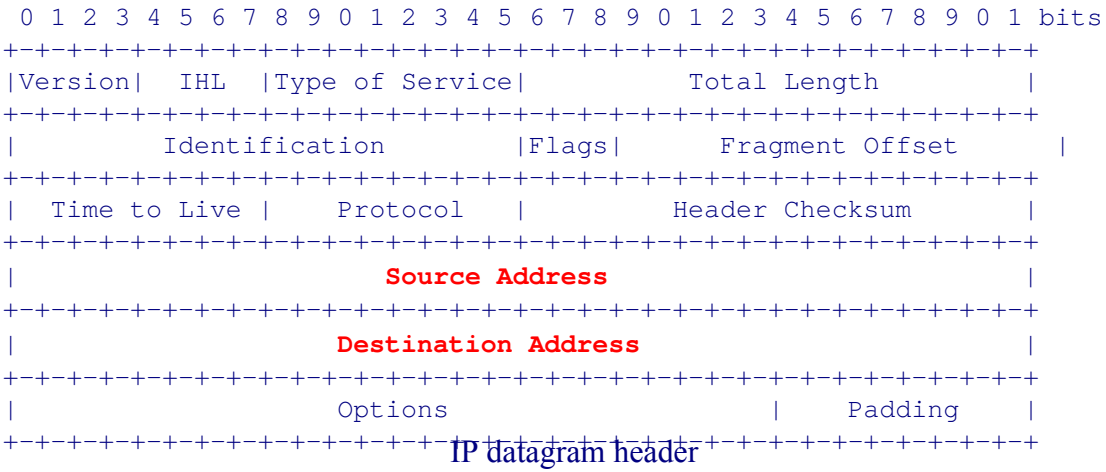
- IP layer service
- **IP addresses**
- Subnetting
- Routing tables
- ARP protocol
- IP header
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- NAT
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Unit 2: IP Networks

IP Addresses (RFC 791)



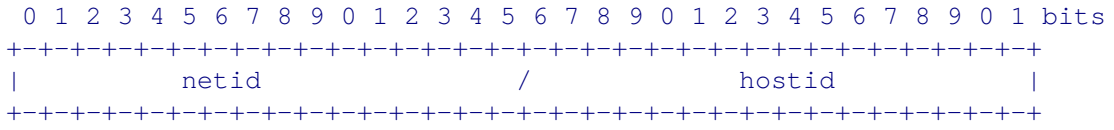
Datagram packet switching



IP datagram header

Unit 2: IP Networks

IP Addresses



- 32 bits (4 bytes).
- Dotted point notation: Four bytes in decimal, e.g. 147.83.24.28
- netid identifies the network.
- hostid identifies the host within the network.
- An IP address identifies an *interface*: an attachment point to the network.
- All IP addresses in Internet must be different. To achieve this goal, Internet Assigned Numbers Authority, IANA (http://www.iana.net) assign address blocs to Regional Internet Registries, RIR:
  - RIPE: Europe, http://www.ripe.net.
  - ARIN: USA, http://www.arin.net.
  - APNIC: ASIA http://www.apnic.net.
  - LACNIC: Latin America, http://www.lacnic.net.
  - AFRINIC: Afica, http://www.afrinic.net.
- RIR assign addresses to ISPs, and ISPs to their customers.

Unit 2: IP Networks

IP Addresses - Classes

- The highest bits identify the class.
- The number of IP bits of netid/hostid varies in classes A/B/C.
- D Class is for multicast addresses (e.g. 224.0.0.2: "all routers")
- E Class are reserved addresses.

Classe	netid (bytes)	hostid (bytes)	Codification	range
A	1	3	0xxxx···x	0.0.0.0 ∼ 127.255.255.255
B	2	2	10xxx···x	128.0.0.0 ∼ 191.255.255.255
C	3	1	110xx···x	192.0.0.0 ∼ 223.255.255.255
D	-	-	1110x···x	224.0.0.0 ∼ 239.255.255.255
E	-	-	1111x···x	240.0.0.0 ∼ 255.255.255.255

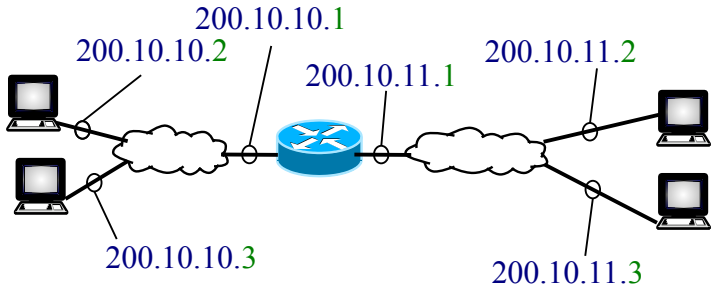
Unit 2: IP Networks

IP Addresses – Special Addresses

- Special addresses cannot be used for a physical interface.
- Each network has two special addresses: network and broadcast addresses.

netid	hostid	Meaning
xxx	all '0'	Identifies a network. It is used in routing tables.
xxx	all '1'	Broadcast in the net. xxx.
all '0'	all '0'	Identifies "this host" in "this net.". Used as source address in configuration protocols, e.g. DHCP.
all '1'	all '1'	broadcast in "this net.". Used as destination address in configuration protocols, e.g. DHCP.
127	xxx	host loopback: interprocess communication with TCP/IP.

- Example:



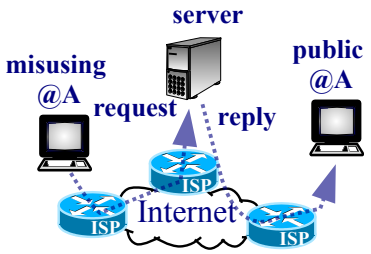
Unit 2: IP Networks

IP Addresses – Private Addresses (RFC 1918)

- Most commercial OSs include the TCP/IP stack.
- TCP/IP is used to network many kind of electronic devices:



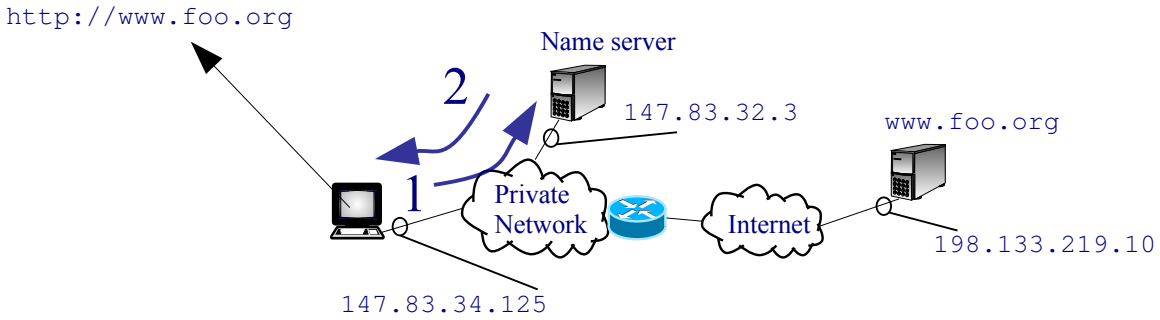
- Addresses assigned to RIRs by IANA are called *public, global* or *registered*.
- What if we arbitrarily assign a registered address to a host?
  - It may be filtered by our ISP or cause trouble to the right host using that address.
- **Private addresses** has been reserved for devices not using public addresses. These addresses are not assigned to any RIR (are not unique). There are addresses in each class:
  - 1 class A network: 10.0.0.0
  - 16 class B networks: 172.16.0.0 ~ 172.31.0.0
  - 256 class C networks: 192.168.0.0 ~ 192.168.255.0



Unit 2: IP Networks

DNS – Protocol (EXPLAINED IN DETAIL IN UNIT 5)

- Client-server paradigm
- Short messages uses UDP.
- well-known port: 53



1 DNS Request

```
18:36:00.322370 IP (proto: UDP) 147.83.34.125.1333 >
      147.83.32.3.53:  53040+ A? www.foo.org. (31)
```

2 DNS Reply

```
18:36:00.323080 IP (proto: UDP) 147.83.32.3.53 > 147.83.34.125.1333:
      53040 1/2/2 www.foo.org. A 198.133.219.10 (115)
```

Unit 2: IP Networks

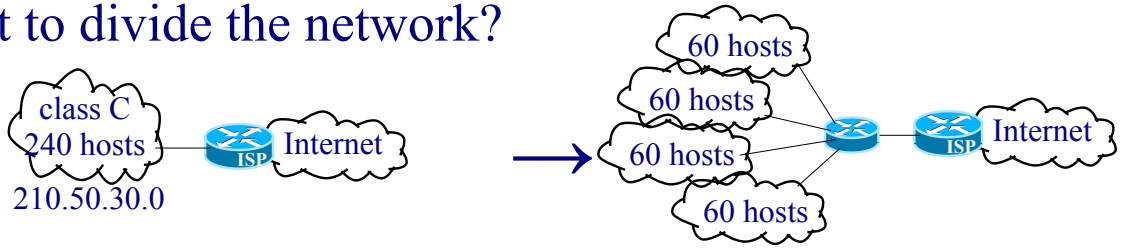
Outline

- IP layer service
- IP addresses
- **Subnetting**
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Unit 2: IP Networks

Subnetting (RFC 950)

- Initially the netid was given by the address class: A with 2^{24} addresses, B with 2^{16} addresses and C with 2^{8} addresses.
- What if we want to divide the network?



- Subnetting allows adding bits from the hostid to the netid (called subnetid bits).
- Example: For the ISP the network prefix is 24 bits. For the internal router the network prefix is 26 bits. The 2 extra bits allows 4 "subnetworks".
- A **mask** is used to identify the size of the netid+subnetid prefix.
- Mask notations:
  - dotted, as 255.255.255.192
  - giving the mask length (number of bits) as 210.50.30.0/26

Unit 2: IP Networks

IP Addresses – Subnetting Example

- We want to subnet the address 210.50.30.0/24 in 4 subnets



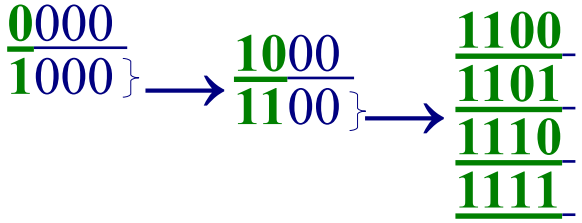
B = 210.50.30

subnet	subnetid	IP net. addr.	range	broadcast	available
S1	00	B.0/26	B.0 ∼ B.63	B.63	$2^6 - 2 = 62$
S2	01	B.64/26	B.64 ∼ B.127	B.127	$2^6 - 2 = 62$
S3	10	B.128/26	B.128 ∼ B.191	B.191	$2^6 - 2 = 62$
S4	11	B.192/26	B.192 ∼ B.255	B.255	$2^6 - 2 = 62$

Unit 2: IP Networks

IP Addresses – Variable Length Subnet Mask (VLSM)

- Subnetworks of different sizes.
- Example, subnetting a class C address:
  - We have 1 byte for subnetid + hostid.
  - subnetid is green, chosen subnets addresses are underlined.



subnet	subnetid	IP net. addr.	range	broadcast	available
S1	0	B.0/25	B.0 ∼ B.127	B.127	$2^7 - 2 = 126$
S2	10	B.128/26	B.128 ∼ B.191	B.191	$2^6 - 2 = 62$
S3	1100	B.192/28	B.192 ∼ B.207	B.207	$2^4 - 2 = 14$
S4	1101	B.208/28	B.208 ∼ B.223	B.223	$2^4 - 2 = 14$
S5	1110	B.224/28	B.224 ∼ B.239	B.239	$2^4 - 2 = 14$
S6	1111	B.240/28	B.240 ∼ B.255	B.255	$2^4 - 2 = 14$

Unit 2: IP Networks

IP Addresses – Classless Inter-Domain Routing, CIDR (RFC 1519)

- Initially, Internet backbone routing tables did not use masks: netid was derived from the IP address class.
- When the number of networks in Internet started growing exponentially, routing tables size started exploding.
- In order to reduce routing tables size, CIDR proposed a "rational" geographical-based distribution of IP addresses to be able to "aggegate routes", and use masks instead of classes.
- Aggregation example:

  200.1.10.0/24
  200.1.11.0/24 ⟶ 200.1.10.0/23

- The term summarization is normally used when aggregation is done at a class boundary (e.g. a groups of subnets is summarized with their classful base address).
- NOTE: Aggregation cannot be done arbitrarily, otherwise the whole routing table could be aggregated in the default route 0.0.0.0/0. E.g. in BGP are specified which ranges can be aggregated, in RIP it is used summarization.

Unit 2: IP Networks

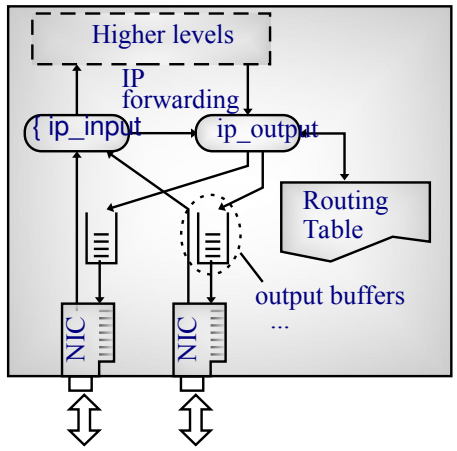
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Unit 2: IP Networks

Routing Table

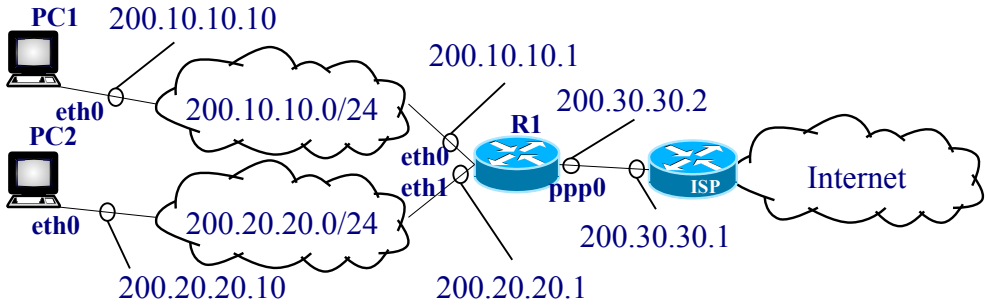
- ip_output() kernel function consults the routing table for each datagram.
- Routing can be:
  - Direct: The destination is directly connected to an interface.
  - Indirect: Otherwise. In this case, the datagram is sent to a router.
- Default route: Is an entry where to send all datagrams with a destination address to a network not present in the routing table. The default route address is 0.0.0.0/0.
- Hosts routing tables usually have two entries: The network where they are connected and a default route.



Basic router architecture

Unit 2: IP Networks

Routing Table – Unix Example



known destinations

how to reach the destinations

PC1 routing table:

Destination	Genmask	Gateway	Iface
200.10.10.0	255.255.255.0	0.0.0.0	eth0
0.0.0.0	0.0.0.0	200.10.10.1	eth0

PC2 routing table:

Destination	Genmask	Gateway	Iface
200.20.20.0	255.255.255.0	0.0.0.0	eth0
0.0.0.0	0.0.0.0	200.20.20.1	eth0

R1 routing table:

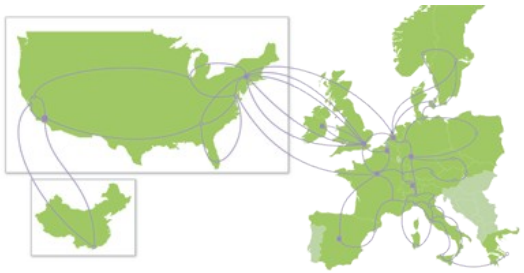
Destination	Genmask	Gateway	Iface
200.10.10.0	255.255.255.0	0.0.0.0	eth0
200.20.20.0	255.255.255.0	0.0.0.0	eth1
200.30.30.1	255.255.255.255	0.0.0.0	ppp0
0.0.0.0	0.0.0.0	200.30.30.1	ppp0

Unit 2: IP Networks

Routing Table – Tiscali ISP, CISCO 7200 Router

- Telnet to route-server.ip.tiscali.net (see http://www.bgp4.net server list)

```
+------------------------------------------------------------------+
|           TISCALI International Network - Route Monitor          |
|                            (AS3257)                              |
|   This system is solely for internet operational purposes. Any   |
|   misuse is strictly prohibited. All connections to this router  |
|   are logged.                                                    |
|   This server provides a view on the TISCALI routing table that  |
|   is used in Frankfurt/Germany. If you are interested in other   |
|   regions of the backbone check out http://www.ip.tiscali.net/lg |
|   Please report problems to noc@tiscali.net                      |
+------------------------------------------------------------------+
```



Tiscali Network Map
http://www.tiscali.net

```
route-server.ip.tiscali.net> show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 213.200.64.93 to network 0.0.0.0
B       85.27.76.0/22 [20/10] via 213.200.64.93, 4w2d
B       85.196.154.0/24 [20/10] via 213.200.64.93, 1d09h
B       85.158.216.0/21 [20/10] via 213.200.64.93, 2w6d
B       85.193.136.0/22 [20/10] via 213.200.64.93, 3d08h
B       85.121.48.0/21 [20/0] via 213.200.64.93, 1w4d
B       85.187.201.0/24 [20/10] via 213.200.64.93, 4d19h
B       85.114.0.0/20 [20/10] via 213.200.64.93, 1w5d
B       85.119.16.0/24 [20/10] via 213.200.64.93, 4w0d
B       85.119.16.0/21 [20/10] via 213.200.64.93, 4w0d
B       85.105.0.0/17 [20/10] via 213.200.64.93, 4w2d
B       85.93.52.0/24 [20/10] via 213.200.64.93, 4w0d
...
```

thousands of entries

Unit 2: IP Networks

Routing Table – Datagram Delivery Algorithm

- 1. Check if the device itself is the destination:

```
if(Datagram Destination == address of any of the interfaces) {
    send the datagram to upper layers
}
```

- 2. Consult the routing table:

```
for each routing table entry ordered from longest to shortest mask
  (Longest Prefix Match) {
    if((Datagram Destination IP address & mask) == Destination
      table entry) {
        return (gateway, interface) ;
    }
```

- 3. Forward the datagram

```
if(it is a direct routing) {
    send the datagram to the Datagram Destination IP address
} else { /* it is an indirect routing */
    send the datagram to the gateway IP address
}
```

Unit 2: IP Networks

Outline

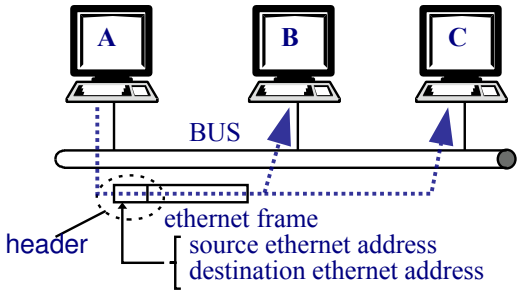
- IP layer service
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Unit 2: IP Networks

Address Resolution Protocol, ARP (RFC 826)

- To send the datagram, IP layer may have to pass a "physical address" to the NIC driver. Physical addresses are also called MAC or hardware addresses.
- ARP translate IP addresses to "physical addresses" (used by the physical network).
- If needed, IP calls ARP module to obtain the "physical addresses" before the NIC driver call.

- Ethernet example:



Unit 2: IP Networks

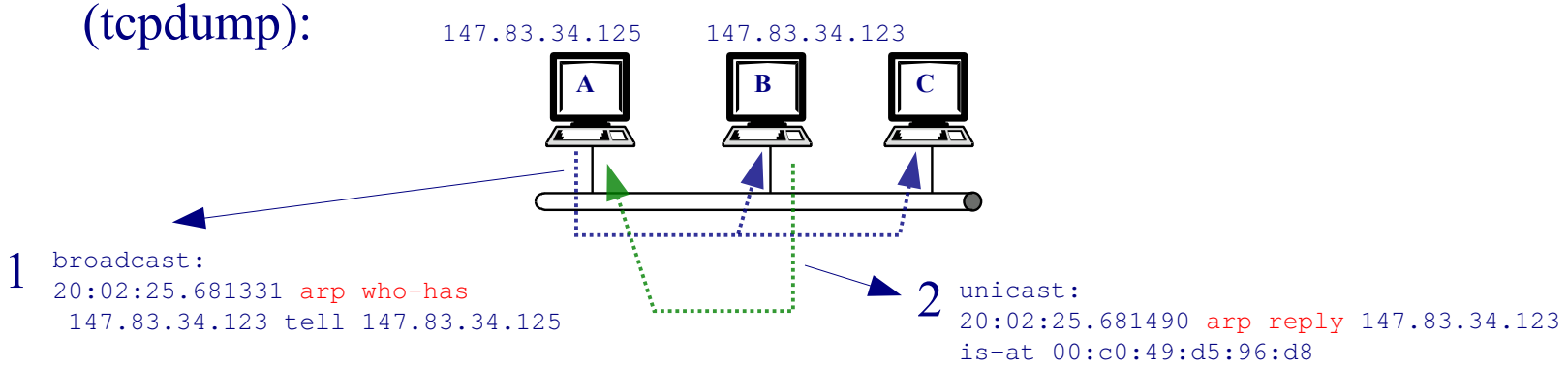
Address Resolution Protocol, messages

- When IP calls ARP:
  - If ARP table has the requested address, it is returned,
  - otherwise:
    - IP stores the datagram in a temporal buffer, and a resolution protocol is triggered.
    - IP initiates a timeout and starts forwarding the next datagram in the transmission queue.
    - If the timeout triggers before resolution, the datagram is removed.
    - If ARP returns the requested address, IP calls the driver with it.
- ARP resolution in an ethernet network (broadcast network):
  - A broadcast "ARP Request" message is sent indicating the IP address.
  - The station having the requested IP address sends a unicast "ARP Reply", and stores the requesting address in the ARP table.
  - Upon receiving the "ARP Reply", the requesting station return the IP call with it.
  - ARP entries have a timeout refreshed each time a match occurs.

Unit 2: IP Networks

Address Resolution Protocol, messages - Example

- ARP messages (tcpdump):



- ARP tables:

```
A> /sbin/arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
147.83.34.123            ether   00:c0:49:d5:96:d8   C                     eth0

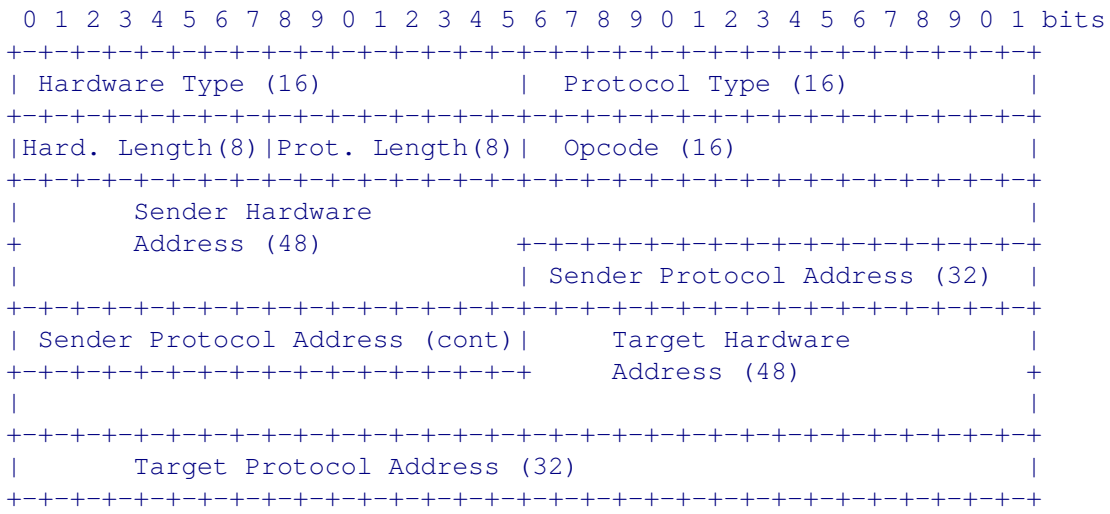
B> /sbin/arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
147.83.34.125            ether   00:14:F1:CC:59:00   C                     eth0
```

"Completed" flag

Unit 2: IP Networks

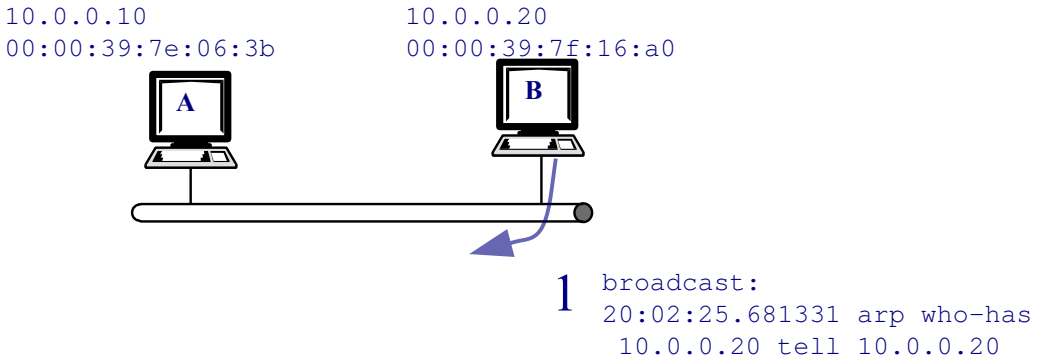
Address Resolution Protocol – Message format (ethernet)

- ARP messages are encapsulated directly in a data-link frame.



Unit 2: IP Networks

Address Resolution Protocol – Gratuitous ARP



- Goals:
  - Detect duplicated IP addresses.
  - Update MAC addresses in ARP tables after an IP or NIC change.

Unit 2: IP Networks

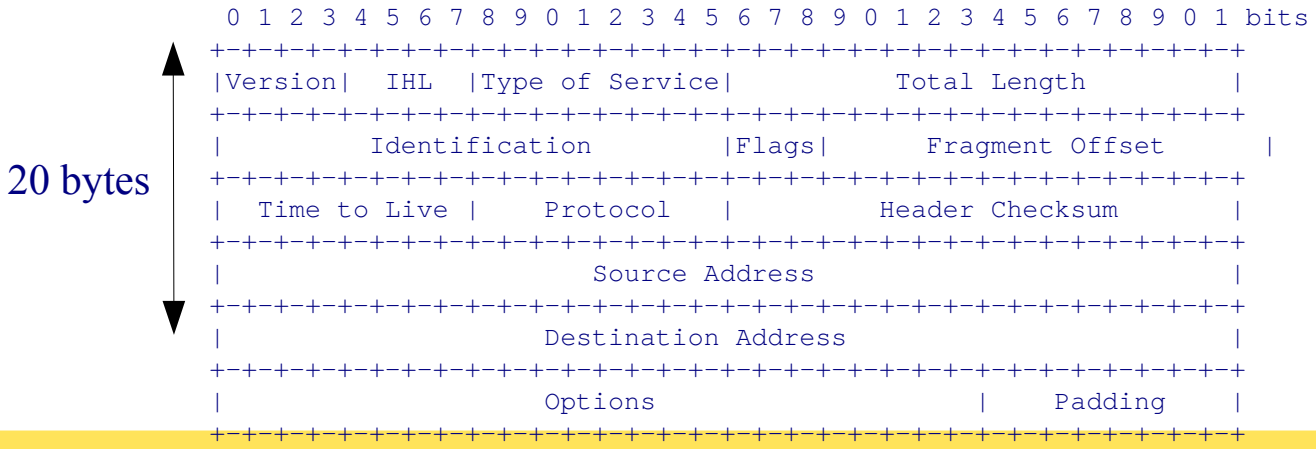
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Unit 2: IP Networks

IP Header (RFC 791)

- Version: 4
- IP Header Length (IHL): Header size in 32 bit words.
- Type of Service: (ToS): *xxxdtrc0*.
- Total Length: Datagram size in bytes.
- Identification/Flags/Fragment Offset: used in fragmentation.
- Time to Live (TTL): if(--TTL==0) { discard ; }.
- Protocol: Encapsulated protocol (/etc/protocols in unix).
- Header Checksum: Header error detection.
- Source and Destination Addresses: End nodes addresses.
- Options: Rcord Route, Loose Source Routing, Strict Source Routing.



Unit 2: IP Networks

IP Fragmentation

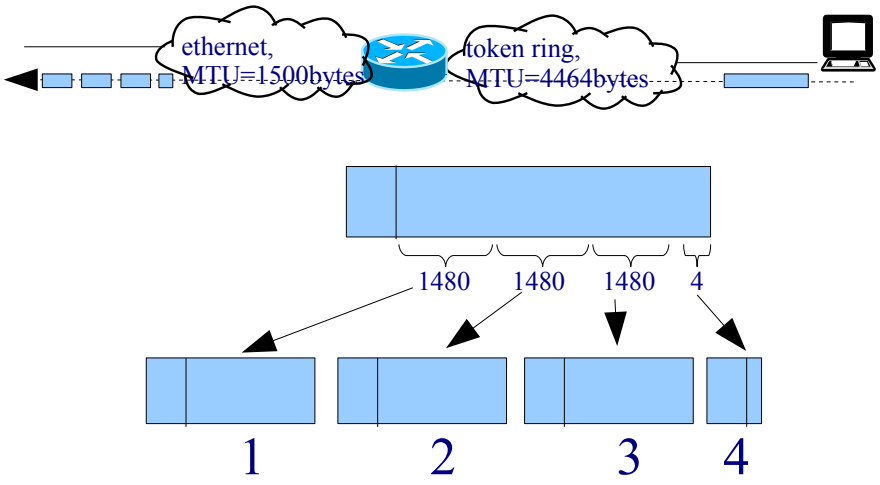
- Fragmentation may occur:
  - Router: Fragmentation may be needed when two networks with different *Maximum Transfer Unit* (MTU) are connected.
  - Host: Fragmentation may be needed using UDP. TCP segments are ≤ MTU.
- Datagrams are reconstructed at the destination.
- Fields:
  - Identification (16 bits): identify fragments from the same datagram.
  - Flags (3 bits):
    - D, don't fragment. Used in MTU path discovery
    - M, More fragments: Set to 0 only in the last fragment
  - Offset (13 bits): Position of the fragment first byte in the original datagram in 8 byte words (indexed at 0).



Unit 2: IP Networks

IP Fragmentation - Example

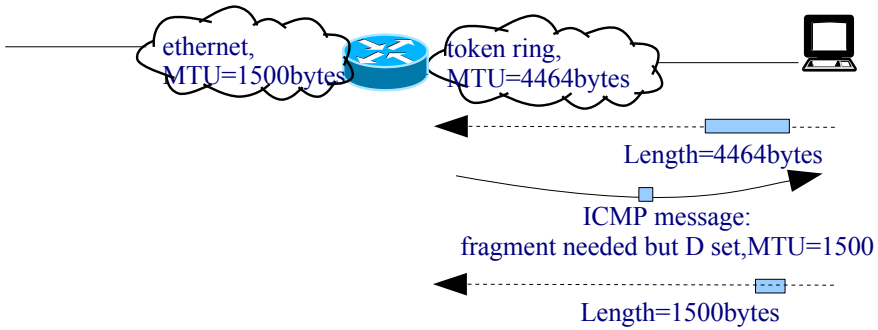
- Original datagram = 4464 bytes (4Mbps Token Ring): 20 header + 4444 payload.
- Fragment size = $\left\lfloor \frac{1500-20}{8} \right\rfloor = 185$ 8-byte-words (1480 bytes)
  - 1st fragment: offset = 0 , M = 1. 0~1479 payload bytes.
  - 2nd fragment: offset = 185, M = 1. 1480~2959 payload bytes.
  - 3rd fragment: offset = 370, M = 1 . 2960~4439 payload bytes.
  - 4th fragment: offset = 555, M = 0 . 4440~4443 payload bytes.



Unit 2: IP Networks

MTU Path Discovery

- Used in modern TCP implementations.
- TCP by default chooses the maximum segment size, to avoid headers overhead (segment efficiency = TCP payload / (TCP payload + Σ TCP,IP,Data-link,Physical headers)
- Goal: avoid fragmentation: The DF flag is set to one, segment size is reduced upon receiving ICMP error message "fragmentation needed but DF flag set"



Unit 2: IP Networks

Outline

- IP layer service
- IP addresses
- Subnetting
- Routing tables
- ARP protocol
- IP header
- **ICMP protocol**
- DHCP protocol
- Routing algorithms
- NAT
- Security in IP

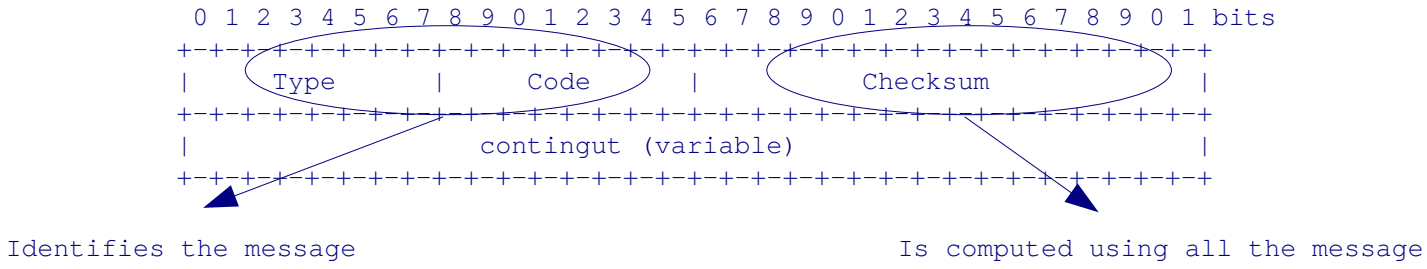
Unit 2: IP Networks

Internet Control Message Protocol, ICMP (RFC 792)

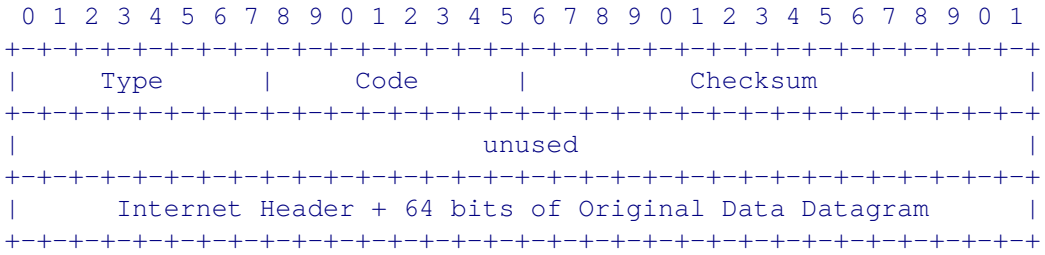
- Used for attention and error messages.
- Can be generated by IP, TCP/UDP, and application layers.
- Are encapsulated into an IP datagram.
- Can be: (i) query, (ii) error.
- An ICMP error message cannot generate another ICMP error message (to avoid loops).

Unit 2: IP Networks

ICMP general format message (RFC 792)



- Query type messages have an identifier field, for request-reply correspondence.
- Error messages have a field where the first 8 bytes of the datagram payload causing the error are copied. These bytes capture the TCP/UDP ports. E.g. Destination Unreachable Message:



Unit 2: IP Networks

Common ICMP messages

Type	Code	query/error	Name	Description
0	0	query	echo reply	Reply an echo request
3	0	error	network unreachable	Network not in the RT.
	1	error	host unreachable	ARP cannot solve the address.
	2	error	protocol unreachable	IP cannot deliver the payload
	3	error	port unreachable	TCP/UDP cannot deliver the payload
	4	error	fragmentation needed and DF set	MTU path discovery
4	0	error	source quench	Sent by a congested router.
5	0	error	redirect for network	When the router send a datagram by the same interface it was received.
8	0	query	echo request	Request for reply
11	0	error	time exceeded, also known as TTL=0 during transit	Sent by a router when `--TTL=0`

Unit 2: IP Networks

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Unit 2: IP Networks

Dynamic Host Configuration Protocol, DHCP (RFC 2131)

- Improves and can interoperate with previous BOOTP protocol.
- Used for automatic network configuration:
  - Assign IP address and mask,
  - Default route,
  - Hostname,
  - DNS domain,
  - Configure DNS servers,
  - etc.
- IP address configuration can be:
  - Dynamic: During a leasing time.
  - Automatic: Unlimited leasing time.
  - Manual: IP addresses are assigned to specific MAC addresses.

Unit 2: IP Networks

DHCP – Protocol Messages (RFC 2131)

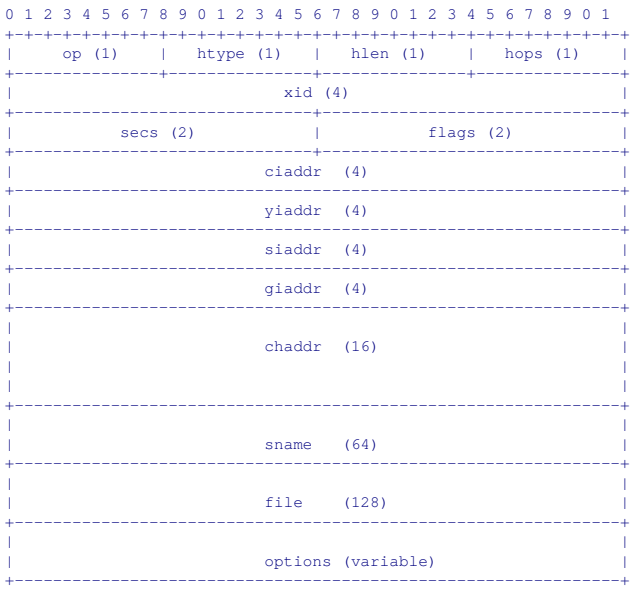
DHCPDISCOVER	-	Client broadcast to locate available servers.
DHCPOFFER	-	Server to client in response to DHCPDISCOVER with offer of configuration parameters.
DHCPREQUEST	-	Client message to servers either (a) requesting offered parameters from one server and implicitly declining offers from all others, (b) confirming correctness of previously allocated address after, e.g., system reboot, or (c) extending the lease on a particular network address.
DHCPACK	-	Server to client with configuration parameters, including committed network address.
DHCPNAK	-	Server to client indicating client's notion of network address is incorrect (e.g., client has moved to new subnet) or client's lease as expired
DHCPDECLINE	-	Client to server indicating network address is already in use.
DHCPRELEASE	-	Client to server relinquishing network address and cancelling remaining lease.
DHCPINFORM	-	Client to server, asking only for local configuration parameters; client already has externally configured network address.

Unit 2: IP Networks

DHCP – Message Fields (RFC 2131)

(informative slide, don't learn the message fields by heart!)

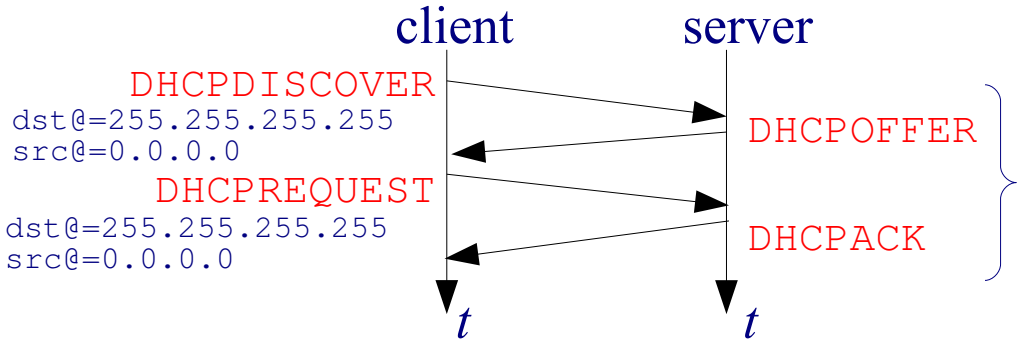
FIELD	OCTETS	DESCRIPTION
op	1	Message op code / message type. 1 = BOOTREQUEST, 2 = BOOTREPLY.
htype	1	Hardware address type.
hlen	1	Hardware address length.
hops	1	Client sets to zero, optionally used by relay agents when booting via a relay agent.
xid	4	Transaction ID, a random number chosen by the client, used by the client and server to associate messages and responses between a client and a server.
secs	2	Filled in by client, seconds elapsed since client began address acquisition or renewal process.
flags	2	Flags.
ciaddr	4	Client IP address; only filled in if client is in BOUND, RENEW or REBINDING state and can respond to ARP requests.
yiaddr	4	'your' (client) IP address. Set by the server in a DHCPOFFER message.
siaddr	4	IP address of next server to use in bootstrap; returned in DHCPOFFER, DHCPACK by server.
giaddr	4	Relay agent IP address, used in booting via a relay agent.
chaddr	16	Client hardware address.
sname	64	Optional server host name, null terminated string.
file	128	Boot file name, null terminated string; "generic" name or null in DHCPDISCOVER, fully qualified directory-path name in DHCPOFFER.
options	var	Optional parameters field.



Unit 2: IP Networks

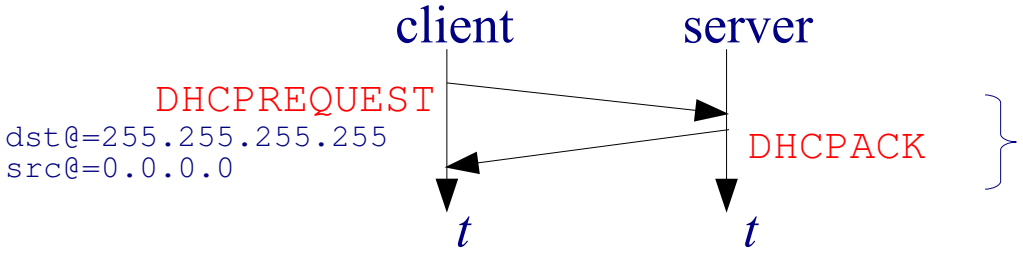
DHCP – Client-server interaction (RFC 2131)

- UDP, server port = 67, client port = 68.



Can be unicast or broadcast, if requested by the client (broadcast flag).

- The client can directly send `DHCPREQUEST`:
  - After rebooting if it remembers and wishes to reuse a previously allocated network address.
  - Extending the lease on a particular network address.



Can be unicast or broadcast, if requested by the client (broadcast flag).

DHCP – Example: tcpdump/dhcpdump capture

```
linux # tcpdump -lenx -s 1500 -i eth0 port bootps or port bootpc | dhcpdump
  TIME: 17:09:24.616312
    IP: 0.0.0.0.68 (00:30:1b:b4:6d:78) > 255.255.255.255.67 (ff:ff:ff:ff:ff:ff)
    OP: 1 (BOOTPREQUEST)
 HTYPE: 1 (Ethernet)
   XID: 181f0139
 FLAGS: 0
CIADDR: 0.0.0.0
YIADDR: 0.0.0.0
SIADDR: 0.0.0.0
GIADDR: 0.0.0.0
CHADDR: 00:30:1b:b4:6d:78:00:00:00:00:00:00:00:00:00:00
OPTION:  53 (  1) DHCP message type         3 (DHCPREQUEST)
OPTION:  57 (  2) Maximum DHCP message size 576
OPTION:  50 (  4) Request IP address        192.168.1.100
OPTION:  51 (  4) IP address leasetime      -1 ()
OPTION:  55 ( 21) Parameter Request List      1 (Subnet mask)
                                              3 (Routers)
                                              6 (DNS server)
                                             12 (Host name)
                                             15 (Domainname)
                                             23 (Default IP TTL)
                                             28 (Broadcast address)
                                             29 (Perform mask discovery)
                                             42 (NTP servers)
                                              9 (LPR server)
                                            119 (Domain Search)
                                            ...
--------------------------------------------------------------------------------
  TIME: 17:09:24.619312
    IP: 192.168.1.1.67 (00:18:39:5d:74:9d) > 192.168.1.100.68 (00:30:1b:b4:6d:78)
    OP: 2 (BOOTPREPLY)
 HTYPE: 1 (Ethernet)
   XID: 181f0139
 FLAGS: 0
CIADDR: 0.0.0.0
YIADDR: 192.168.1.100
SIADDR: 192.168.1.1
GIADDR: 0.0.0.0
CHADDR: 00:30:1b:b4:6d:78:00:00:00:00:00:00:00:00:00:00
OPTION:  53 (  1) DHCP message type         5 (DHCPACK)
OPTION:  54 (  4) Server identifier         192.168.1.1
OPTION:  51 (  4) IP address leasetime      86400 (24h)
OPTION:   1 (  4) Subnet mask               255.255.255.0
OPTION:   3 (  4) Routers                   192.168.1.1
```

Unit 2: IP Networks

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- **Routing algorithms**
- Security in IP

Unit 2: IP Networks

Routing algorithms

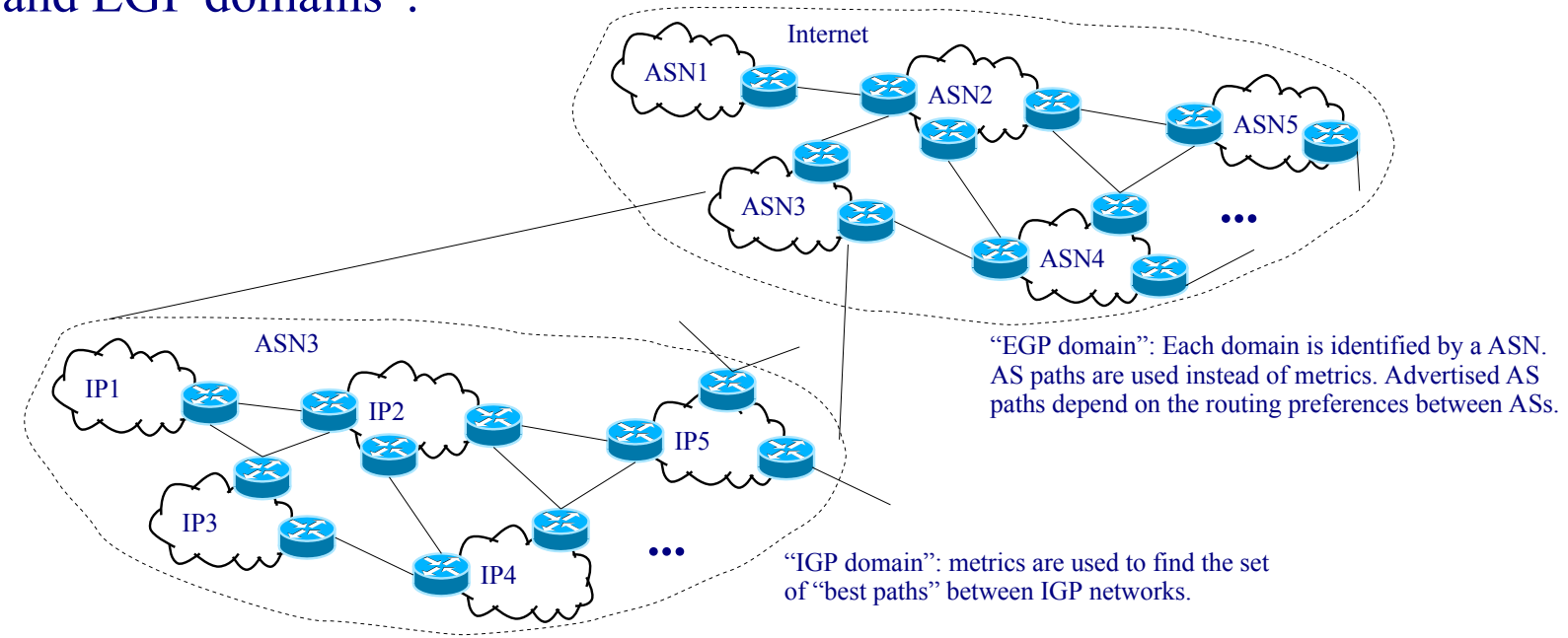
- Objective: add entries to routing tables. Can be:
  - Static: Manual, scripts, DHCP.
  - Dynamic: Automatically update table entries, e.g. when a topology change occurs. This is done by a routing algorithm.

- Internet is organized in Autonomous Systems (AS). In terms of ASs, routing algorithms are classified as:
  - Interior Gateway Protocols (IGPs): Inside the same AS. Examples:
    - RFC standards: RIP, OSPF.
    - Proprietary: CISCO IGRP.
  - Exterior Gateway Protocols (EGPs): Between different ASs. Currently BGPv4.

Unit 2: IP Networks

Routing algorithms - Autonomous Systems (AS)

- AS definition (RFC 1930): "An AS is a connected group of one or more IP prefixes run by one or more network operators which has a SINGLE and CLEARLY DEFINED routing policy".
- Each AS is identified by a 16 bits AS Number (ASN) assigned by IANA.
- ASs facilitate Internet routing by introducing a two-level hierarchy: "IGP and EGP domains".



Unit 2: IP Networks

Routing Information Protocol, RIP (RFC 2453)

- The metric (distance) to a destination is the number of hops (i.e. transmissions) to reach the destination: 1 if the destination is attached to a directly connected network, 2 if 1 additional router is needed ...
- Routers send RIP updates every 30 seconds to the neighbors.
- RIP updates use UDP, src./dst. well-known port = 520, broadcast dst. IP addr.
- RIP updates include destinations and metrics tuples.
- A neighbor is considered down if no RIP messages are seen during 180 seconds.
- Infinite metric is 16.
- Two versions of RIP: Version 2 allows variable masks ans uses the multicast dst. address 224.0.0.9 (all RIPv2 routers).
- This type of routing algorithms, where it is not known the whole topology but the distance to each destination, are known as "distance-vector" or "Bellman-Ford".

Unit 2: IP Networks

Route summarization used by RIP

- All routing algorithms have route aggregation mechanisms that allow reducing the number of routing entries.
- RIP uses a simple aggregation method, referred to as summarization, that consists of aggregating to the "base address with class". For instance, 10.1.0.0/24 and 10.2.0.0/24 are class A addresses, thus are aggregated to 10.0.0.0/8. When the aggregation is done, the aggregated address is sent with metric 1.
- Aggregation is done only when the base address of the network where the update is sent, is different from the aggregated address. Otherwise inconsistencies could occur. For instance, consider the following example:

  10.1.0.0/24 - R1 - 10.2.0.0/24 - R2 - 10.3.0.0/24 - R3 - 10.4.0.0/24

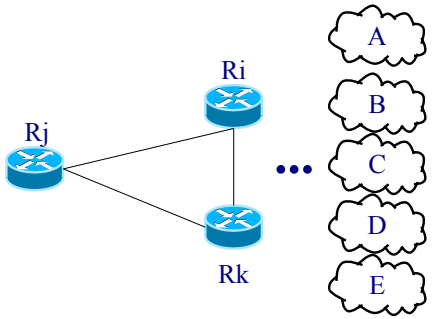
  In this example no router would summarize. Otherwise R1 and R3 would send 10.0.0.0/8, and R2 would not know where 10.1.0.0/24 and 10.4.0.0/24 are.
- Similarly, when the addressing scheme is designed, subnetworks of the same base address must not be partitioned (that is, separated by networks with a different base address). Otherwise routers would summarize, and inconsistencies could appear. For instance, the following addressing would not be valid for RIP with summarization:

  10.1.0.0/24 - R1 – 192.168.0.0/24 - R2 – 192.168.1.0/24 - R3 - 10.4.0.0/24

Unit 2: IP Networks

RIP – Routing Table (RT) Update Example

- Example: When Ri receives an update message from Rj:
  - Increase the message metrics.
  - Add new destinations.
  - Change entries with other routers with larger metrics.
  - Update metrics using Rj's gateway.



D	G	M
A	Rk	4
B	Rj	3
C	Rk	5
D	Rj	2

Ri's RT

D	M
A	1
B	4
C	5
D	1
E	3

Ri receives Rj's update message

D	M
A	2
B	5
C	6
D	2
E	4

Rj's metrics increased

D	G	M
A	Rj	2
B	Rj	5
C	Rk	5
D	Rj	2
E	Rj	4

Ri's RT updated

Unit 2: IP Networks

RIP – Count to Infinity

- Depending on the route update message order, convergence problems may arise:



D	G	M
N1	*	1
N2	*	1
N3	R2	2
N4	R2	3

R1's RT

D	G	M
N1	R1	2
N2	*	1
N3	*	1
N4	R3	2

R2's RT

D	G	M
N1	R2	3
N2	R2	2
N3	*	1
N4	*	1

R3's RT

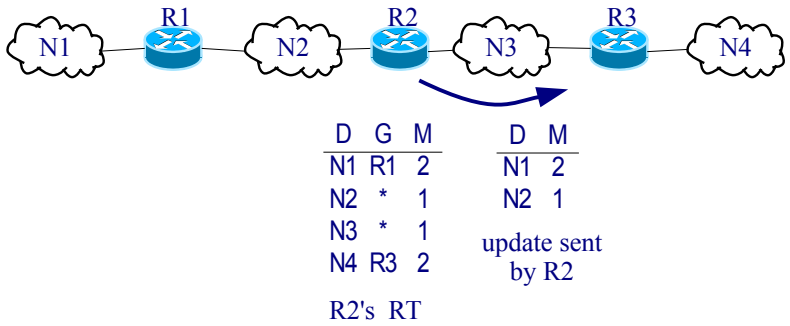
- Evolution of D=N4 entry when R3 fails:

	G	M	R3 fails	G	M	R1 upd	G	M	R2 upd	G	M	R1 upd	G	M	...	G	M
R1:	R2	3	→	R2	3	→	R2	3	→	R2	**5**	→	R2	5	...	R2	**16**
R2:	R3	2	→	R3	**16**	→	R1	**4**	→	R1	4	→	R1	**6**	...	R1	**16**

Unit 2: IP Networks

RIP – Count to Infinity Solutions

- Split horizon: When the router sends the update, removes the entries having a gateway in the interface where the update is sent:



- Split horizon with Poisoned Reverse: Consists of adding the entries having a gateway with M=16.
- Triggered updates: Consists of sending the update before the 30 seconds timer expires when a metric change in the routing table.
- Hold down timer (CISCO): When a route becomes unreachable (metric = 16), the entry is placed in *holddown* during 280 seconds. During this time, the entry is not updated.

Unit 2: IP Networks

Open Shortest Path First, OSPF (RFC 2328)

- IETF standard for high performance IGP routing protocol.
- *Link State* protocol: Routers monitor neighbor routers and networks and send this information to all OSPF routers (*Link State Advertisements*, LSA).
- LSA are encapsulated into IP datagrams with multicast destination address 224.0.0.5, and routed using *flooding*.
- LSA are only sent when changes in the neighborhood occur, or when a LSA Request is received.
- Neighbor routers are monitored using a *hello protocol*.
- OSPF routers maintain a LS database with the information received with LSA. The Shortest Path First algorithm (Dijkstra algorithm) is used to optimal build routing table entries.
- The metric is computed taking into account link bitrates, delays etc.
- The infinite metric is the maximum metric value.
- There is no convergence (count to infinity) problems.

Unit 2: IP Networks

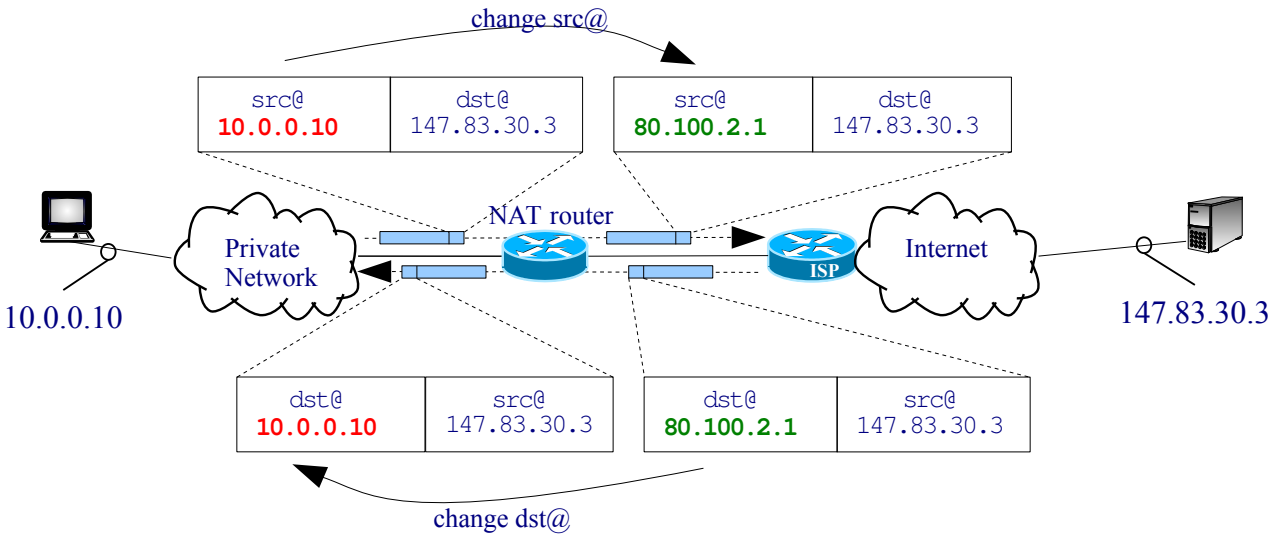
Outline

- IP layer service
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- Routing algorithms
- **NAT**
- Security in IP

Unit 2: IP Networks

Network Address Translation, NAT (RFCs 1631, 2663 3022)

- Typical scenario: Private addresses (internal addresses) are translated to public addresses (external addresses).
- A NAT table is used for address mapping.
- Advantages:
  - Save public addresses.
  - Security.
  - Administration, e.g. changing ISP does not imply changing private network addressing.



Unit 2: IP Networks

NAT – Types of translations

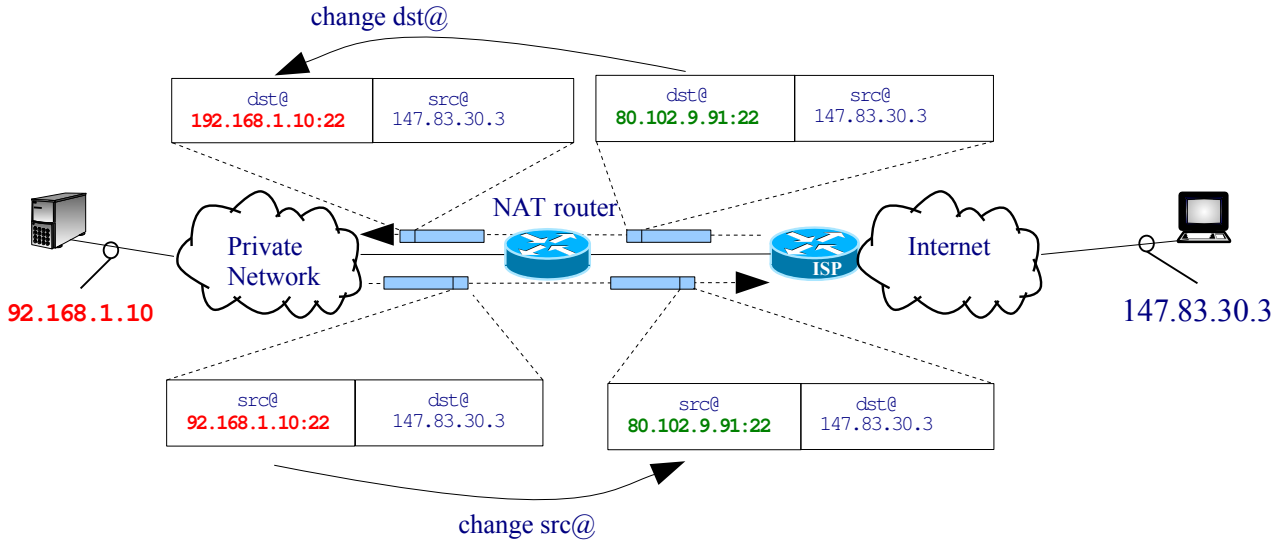
- NOTE: NAT is a technique, not a protocol. Implementations and terminology may change from one manufacturer to another.
- Basic NAT:
  - A different external address is used for each internal address →a different public IP address is needed for each hosts accessing Internet.
  - Each NAT table entry has the tuple: (internal address, external address).
  - Each host requires one NAT table entry.
- Port and Address Translation, PAT:
  - The same external address can be used for each internal address →a unique public IP address can be used for all hosts accessing Internet.
  - Each NAT table entry has the tuple: (int. address/port, ext. address/port)
  - For ICMP echo-req/echo-rep the query-id is used instead of port.
  - Each connection requires one NAT table entry.
- The NAT table entries can be:
  - Static: Manually added.
  - Dynamic:
    - Entries are automatically added when an internal connection is initiated.
    - External addresses are chosen from a pool. Table entries have a timeout.

Unit 2: IP Networks

DNAT

- What if we want external connections to internal servers? (DNAT in linux-iptables terminology).
- The address translation is exactly the same as NAT, but, the connection is initiated from an external client.
- Typically, some static configuration is needed to configure the server IP/port.

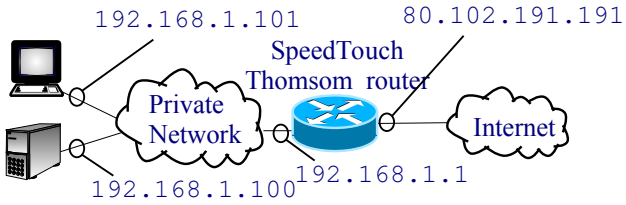
```
Static entry in the NAT router:
Inside-address:Port  Outside-address:Port
192.168.1.10:22      80.102.9.91:22
```



Unit 2: IP Networks

NAT – ADSL commercial router example

- NAT outgoing packets to 80.102.191.191
- DNAT incoming packets, port 22 (ssh) to 192.168.1.100




```
linux # telnet 192.168.1.1
Trying 192.168.0.1...
Connected to 192.168.1.1.
=>nat
[nat]=>list
Indx Prot Inside-address:Port  Outside-address:Port  Foreign-address:Port Flgs   Expir  State  Control
   2    6   192.168.1.100:22     80.102.191.191:22           0.0.0.0:0     instance
   6    6   192.168.1.101:1420   80.102.191.191:10079    83.60.122.22:45730   1   14m48      1
  11    6   192.168.1.101:1337   80.102.191.191:10060   85.56.136.231:16000   1   14m30      1
  12    6   192.168.1.101:1402   80.102.191.191:10064    82.159.8.187:1755    1     14s      5
  ...
```

DNAT

SNAT

Unit 2: IP Networks

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Unit 2: IP Networks

Security in IP

- Goals:
  - Confidentiality: Who can access.
  - Integrity: Who can modify the data.
  - Availability: Access guarantee.
- Vulnerabilities:
  - Technological: Protocols (e.g. ftp and telnet send messages in “clear text”) and networking devices (routers...)
  - Configuration: Servers, passwords, ...
  - Missing security policies: Secure servers, encryption, firewalls, ...

Unit 2: IP Networks

Security in IP – Attacks

- Reconnaissance: Previous to an attack.
  - Available IP addresses.
  - Available servers and ports.
  - Types of OSs, versions, devices...
  - Eavesdropping
- Access: Unauthorized access to an account or service.
- Denial of Service: Disables or corrupts networks, systems, or services.
- Viruses, worms , trojan horses...: Malicious software that replicate itself.

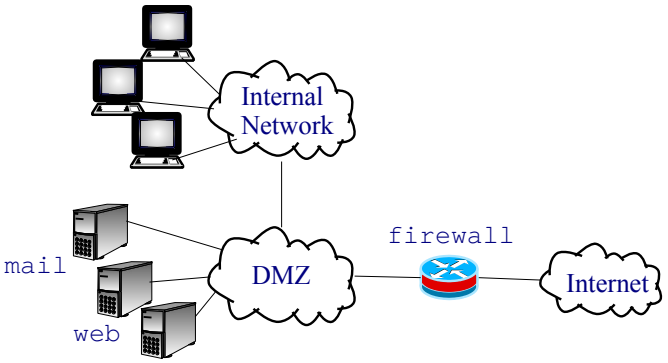
Security in IP – Basic Solutions

- Firewalls.
- Virtual Private Networks (VPN).

Unit 2: IP Networks

Security in IP – Firewalls

- Firewall: System or group of systems that enforces an access control policy to a network.
- There are many firewall types: From simple packet filtering based on IP/TCP/UDP header rules, to state-full connection tracking and application-based filtering, defense against network attacks, ...



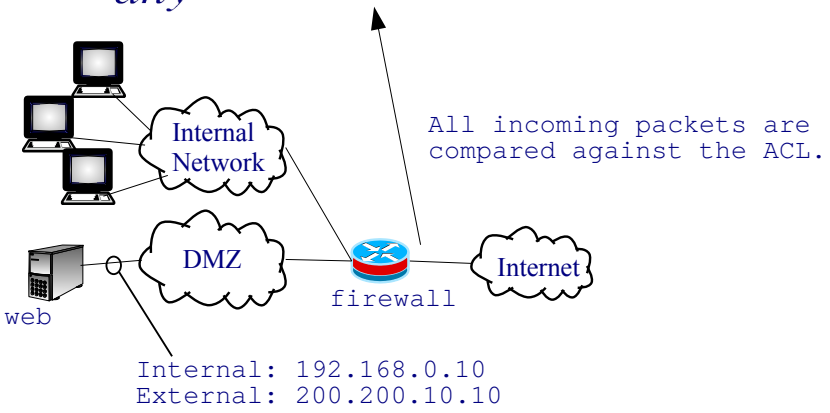
DMZ: Demilitarized Zone. Contains servers exposed to the Internet.

Unit 2: IP Networks

Security in IP – Basic Firewall Configuration

- NAT
- Access Control List, ACL
- Example: assume the following ACL applied to all packets entering the router from the Internet. It allows the clients in the internal network to access the Internet, clients in the Internet to access the web server in the DMZ, and ICMP messages.

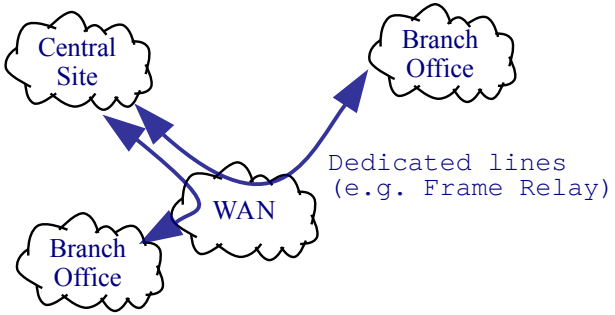
Protocol	IP-src	IP-dst	Port-src	Port-dst	Action
TCP	*any*	200.200.10.10/32	*any*	80	*accept*
TCP	*any*	*any*	< 1024	≥ 1024	*accept*
ICMP	*any*	*any*	–	–	*accept*
IP	*any*	*any*	–	–	*deny*



Unit 2: IP Networks

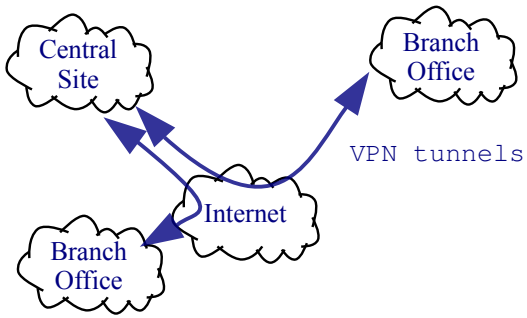
Security in IP – Virtual Private Network, VPN

- Provides connectivity for remote users over a public infrastructure, as they would have over a private network.



Conventional Private Network

- More cost.
- Less flexible.
- WAN management.



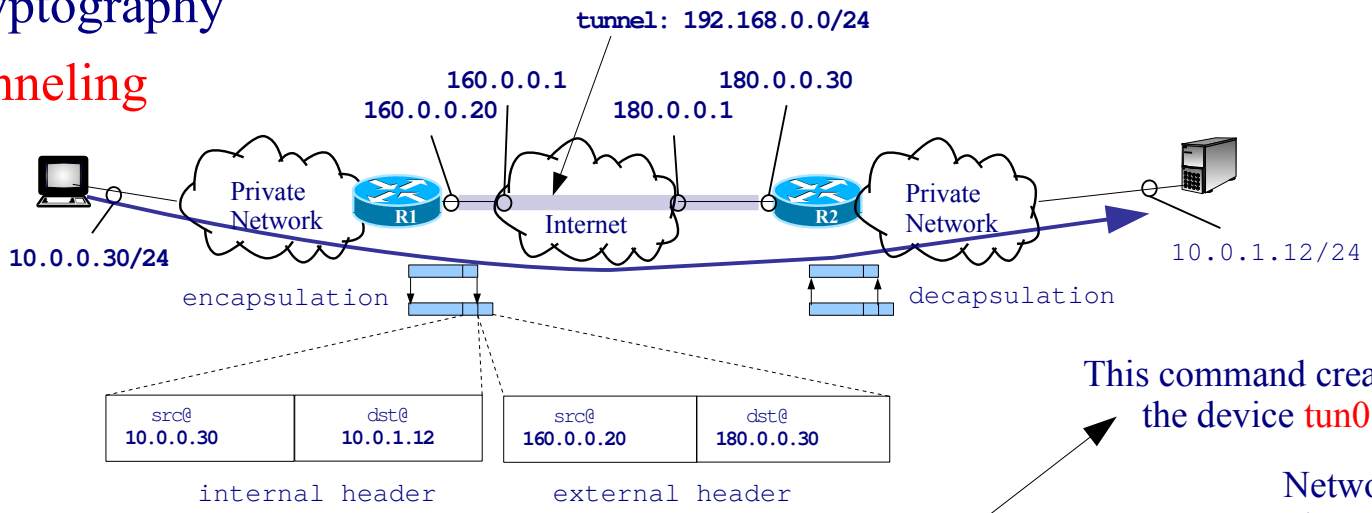
VPN

- Less cost.
- More flexible.
- Simple management.
- Internet availability.

Unit 2: IP Networks

Security in IP – VPN Security

- Authentication
- Cryptography
- Tunneling



This command creates the device tun0

Network reachable through the tunnel

Network of the tunnel

```
Example: creating a tunnel in linux:
R1# ip tunnel add tun0 mode ipip remote  180.0.0.30 local 160.0.0.20
```

Destination	Gateway	Genmask	Iface
10.0.0.0	0.0.0.0	255.255.255.0	eth0
160.0.0.1	0.0.0.0	255.255.255.255	ppp0
0.0.0.0	160.0.0.1	0.0.0.0	ppp0
192.168.0.0	0.0.0.0	255.255.255.0	tun0
10.0.1.0	192.168.0.2	255.255.255.0	tun0

R1 Routing Table

Destination	Gateway	Genmask	Iface
10.0.1.0	0.0.0.0	255.255.255.0	eth0
180.0.0.1	0.0.0.0	255.255.255.255	ppp0
0.0.0.0	180.0.0.1	0.0.0.0	ppp0
192.168.0.0	0.0.0.0	255.255.255.0	tun0
10.0.0.0	192.168.0.1	255.255.255.0	tun0

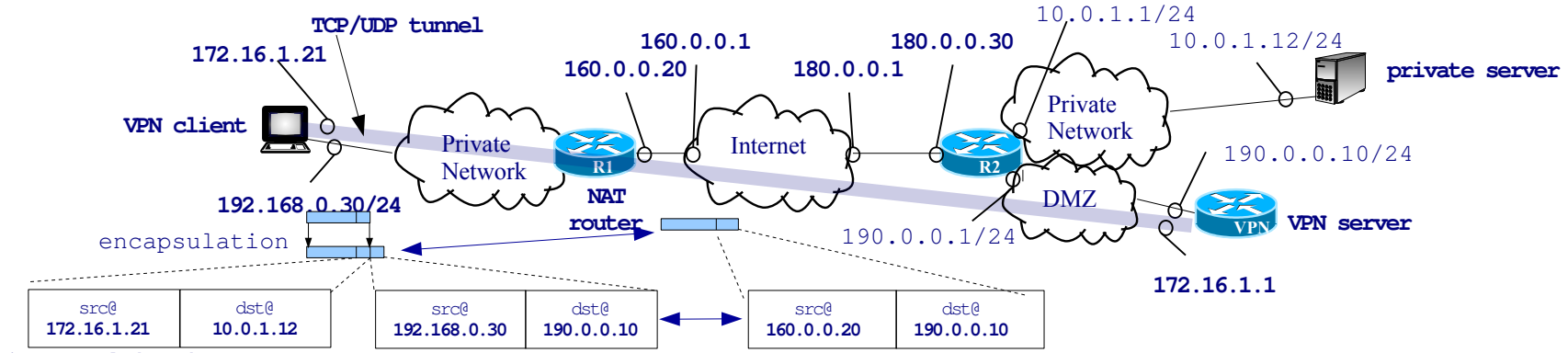
R2 Routing Table

Unit 2: IP Networks

- **TCP/UDP tunnels:**

  Typically used in VPNs to be able to cross NAT routers

Example: openvpn client



internal header (datagrams sent inside the TCP/UDP connection)

external header (adresses of the TCP/UDP socket)

Client routing table

Destination	Gateway	Genmask	Iface
192.168.0.0	0.0.0.0	255.255.255.0	eth0
0.0.0.0	192.168.0.1	0.0.0.0	eth0
172.16.1.1	0.0.0.0	255.255.255.255	tun0
10.0.1.0	172.16.1.1	255.255.255.0	tun0

Network reachable through the tunnel

Private server routing table

Destination	Gateway	Genmask	Iface
10.0.1.0	0.0.0.0	255.255.255.0	eth0
0.0.0.0	10.0.1.1	0.0.0.0	eth0

VPN server routing table

Other VPN clients

Destination	Gateway	Genmask	Iface
190.0.0.0	0.0.0.0	255.255.255.0	eth0
172.16.1.21	0.0.0.0	255.255.255.255	tun0
172.16.1.23	0.0.0.0	255.255.255.255	tun3
...			
0.0.0.0	190.0.0.1	0.0.0.0	eth0

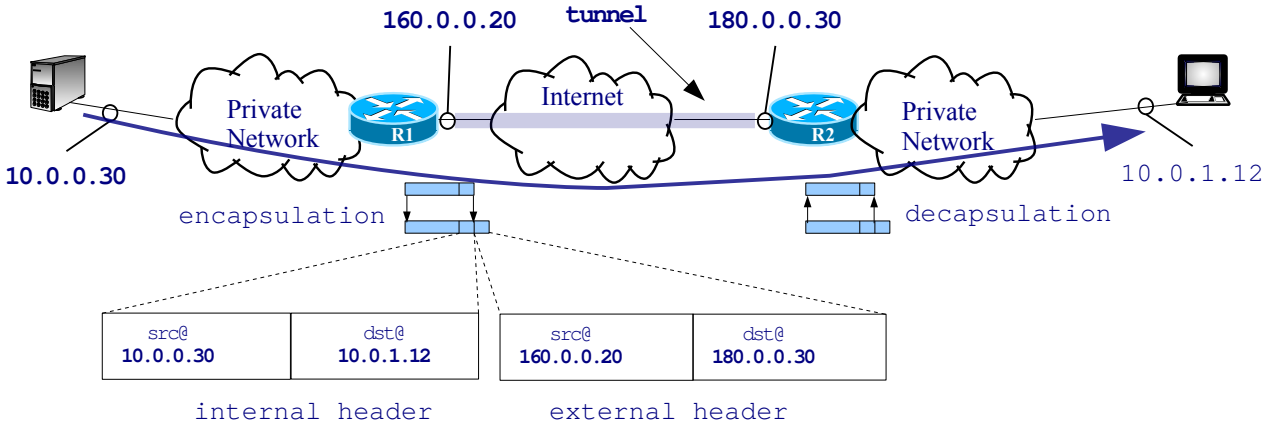
R2 routing table

Destination	Gateway	Genmask	Iface
10.0.1.0	0.0.0.0	255.255.255.0	eth0
190.0.0.0	0.0.0.0	255.255.255.0	eth1
172.16.1.0	190.0.0.10	255.255.255.0	eth1
180.0.0.1	0.0.0.0	255.255.255.255	ppp0
0.0.0.0	180.0.0.1	0.0.0.0	ppp0

VPN network

Unit 2: IP Networks

- **Tunneling Potential Problems**
  - Fragmentation inside the tunnel will use the external header, thus, the exit router of the tunnel may reassemble fragmented datagrams.
  - ICMP messages sent inside the tunnel are addressed to the tunnel entry.
  - MTU path discovery may fail.
  - Solution: the router entry maintains a "tunnel state", e.g. the tunnel MTU, and generate ICMP messages that would be generated inside the tunnel. Furthermore, the tunnel entry router typically fragment the datagrams, if needed, before encapsulation, to avoid the exit router having to reassemble fragmented datagrams.



Unit 2: IP Networks

- **Other types of tunnels:**
  - IP over IP (RFC 2003): Basic encapsulation.
  - Generic Routing Encapsulation, GRE (RFC 1701): There is an additional GRE header: allows encapsulating other protocols (not only IP).
  - Point-to-Point Tunneling Protocol, PPTP (RFC 2637): Add the ppp functionalities.
  - IPsec (RFC 2401): Standards to introduce authentication and encryption and tunneling to IP layer.